

Answer **all** questions.

1. Sioned reacted sulfuric acid with zinc oxide (ZnO).
She followed the method below to form zinc sulfate.

1. Pour 50 cm³ of 1.0 mol/dm³ sulfuric acid into a conical flask.
2. Heat the sulfuric acid to 50°C.
3. Add 5 g of zinc oxide to the warmed acid and stir.
4. Repeat step 3 until the zinc oxide is in excess.

- (a) The main hazard in this experiment is that 1.0 mol/dm³ sulfuric acid is an irritant.

- (i) Describe **one** risk associated with using sulfuric acid. [1]

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- (ii) State a suitable control measure Sioned should carry out to reduce this risk. [1]

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- (b) Circle the correct formula in each bracket to complete the symbol equation for this reaction. [2]



- (c) State **two** additional steps that Sioned needs to carry out to obtain pure zinc sulfate crystals. [2]

1.

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2.

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- (d) Robert has calculated the relative formula mass of zinc oxide to be 97.
Use the Periodic Table on page 20 to explain if he is correct.

[2]

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- (a) (i) State the dependent variable in this experiment. [1]

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- (ii) State **two** variables that should be controlled. [2]

1.

2.

- (b) (i) State how the concentration of dye left in the test tube changed during the first 20 minutes. [1]

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- (ii) Explain why this change occurred. [2]

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- (c) One student suggests that the rate of diffusion is constant over the whole 80 minutes. Explain whether you agree. [3]

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- (d) The students repeated the experiment in a water bath at 30°C. **Add another line** to the graph to show the expected results. [2]

